

MpxR Cleaves a Periplasmic Fluorescent Reporter Independently of the Canonical Heat-Shock Regulators

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ABSTRACT

MpxR is a fictional inner-membrane protease predicted to contain six transmembrane helices. We tested whether MpxR cleaves a periplasmic fluorescent reporter and whether this activity requires DnaK, DnaJ, sigma32, or FtsH. Reporter cleavage increased 4.6-fold when wild-type MpxR was expressed, whereas the catalytic Ser201Ala variant was inactive. Depletion of DnaK, DnaJ, sigma32, or FtsH did not reduce cleavage under the tested conditions. MpxR abundance and membrane localization were unchanged across regulator backgrounds. The data identify catalytic MpxR as sufficient for reporter cleavage and provide no evidence that the canonical heat-shock regulators are required.

Keywords: *MpxR, membrane protease, periplasmic reporter, heat-shock regulator, negative result*

INTRODUCTION

Membrane proteases are often discussed alongside cellular stress systems because damaged membrane proteins can trigger broad quality-control responses. However, co-occurrence in the literature does not establish that a particular protease depends on DnaK, DnaJ, sigma32, or FtsH.

This study examined a single reaction: cleavage of a periplasmic fluorescent reporter by MpxR. The heat-shock regulators were included as genetic controls. The primary hypothesis was that catalytic MpxR would increase reporter cleavage; regulator dependence was a secondary question.

MATERIALS AND METHODS

Reporter system

The reporter consisted of a periplasmic fluorescent domain linked to a quencher through a 14-residue cleavage sequence. Cleavage increased fluorescence. MpxR and MpxR-Ser201Ala were expressed from matched low-copy plasmids.

Regulator backgrounds

Conditional depletion strains were used for DnaK, DnaJ, sigma32, and FtsH. Depletion was verified by immunoblotting. Assays were performed during a 45-min window that preserved cell viability and membrane integrity.

Localization and cleavage

Membranes and soluble fractions were separated by ultracentrifugation. Reporter fluorescence was normalized to cell density. Cleavage products were confirmed by immunoblotting. No affinity chromatography, surface plasmon resonance, or phosphorylation assay was performed.

Construct	Catalytic residue	Expected activity
Empty vector	None	Baseline
MpxR-WT	Ser201	Active
MpxR-S201A	Ala201	Inactive control

Table 1. MpxR constructs.

RESULTS

MpxR-dependent reporter cleavage

Wild-type MpxR increased normalized reporter fluorescence 4.6-fold relative to empty vector. A cleavage fragment of the expected apparent size was detected. MpxR-Ser201Ala produced 1.1-fold baseline fluorescence and no detectable fragment.

Heat-shock regulators were dispensable

DnaK, DnaJ, sigma32, and FtsH depletion changed MpxR-dependent fluorescence by less than 12%. None of the four comparisons met the prespecified 20% effect threshold. MpxR membrane abundance was similar in all backgrounds.

Specificity controls

A reporter with a scrambled cleavage sequence was not cleaved. General membrane permeability remained unchanged, excluding nonspecific release of fluorescence.

Background	Relative cleavage	MpxR membrane level
Control	4.6	1.00
DnaK depleted	4.4	0.97
DnaJ depleted	4.2	1.03
sigma32 depleted	4.5	0.95
FtsH depleted	4.1	1.02

Table 2. Regulator depletion did not materially reduce MpxR cleavage.

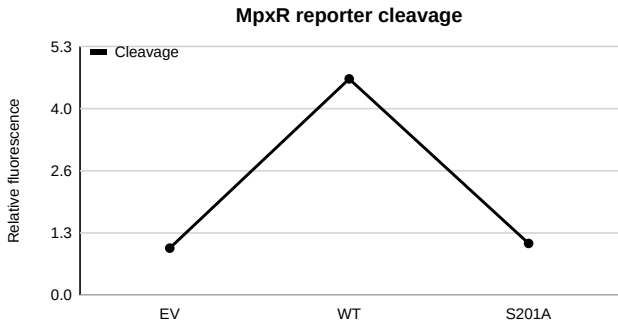


Figure 1. Catalytically active MpxR, but not Ser201Ala, cleaved the reporter.

DISCUSSION

MpxR cleavage depended on its catalytic serine and on the reporter sequence. The tested heat-shock regulators were not required during the assay window. This is a negative genetic result, not evidence that the regulators are absent from membrane quality control in general.

The large reference literature on DnaK, DnaJ, sigma32, and FtsH provides context for why they were tested, but those proteins are not the central subject of this paper.

LIMITATIONS

Short-term depletion may not reveal long-term indirect effects. Only one reporter and one growth condition were tested.

CONCLUSION

Catalytic MpxR was sufficient to cleave the periplasmic reporter. DnaK, DnaJ, sigma32, and FtsH were not required under the tested conditions.

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